

BOUNDARY-SPANNING ACTIVITY AND EMPLOYEE REACTIONS: AN EMPIRICAL STUDY

Robert T. Keller, University of Houston
Andrew D. Szilagyi, University of Houston
Winford E. Holland, University of Houston

ABSTRACT

Boundary-spanning activity was studied in a large manufacturing company through a sample of 192 managers, engineers, and supervisors. Contrary to prior theory and research, this study found boundary-spanning activity unrelated to role conflict or ambiguity and positively related to job satisfaction for the total sample. Boundary-spanning activity was also positively related to a number of job characteristics for the total sample. Marked differences in boundary-spanning activity and its relationships with other variables, however, were found across occupational levels. While managers and engineers generally had boundary-spanning activity related to high levels of job satisfaction and job characteristics, first-level supervisors had boundary-spanning activity related to higher role conflict and lower job satisfaction with opportunities for promotion.

INTRODUCTION

The purpose of the present research was to study the relationships between boundary-spanning activity (BSA) and role and satisfaction variables in a large manufacturing company. In addition, this study also investigated the relationships between BSA and job characteristics in order to gain some insight into the underlying job factors which may be affected by BSA.

BSA, which may be defined as the interpersonal transfer of information across organizational boundaries, has been found to be vital for effective organizational functioning by research into a number of diverse organizations (see literature review in [7]). Thompson's [16] conceptual framework also places considerable importance on boundary-spanning components and BSA for organizational viability. BSA is seen to occur most often at the higher levels of the organization where management is interacting with the environment, while those employees at lower levels in the technical core are usually buffered from environmental factors. Thompson also sees BSA increasing the sphere of action of a person engaging in the activity.

Much of the prior empirical research, however, has found negative effects from BSA. Kahn et al. [6], Organ [9], Organ and Greene [10], and Wall and Adams [17] found boundary spanners to experience such effects as role conflict and tension, and conformity to norms.

The Kahn et al. [6] theory of role dynamics and Thompson's [16] conceptual framework were used in the present study to generate hypotheses regarding BSA. In the Kahn et al. theory perceptions of inconsistent and vague expectations by the role incumbent tend to result in role conflict and ambiguity, and, consequently, coping responses to conflict and ambiguity

are posited to include low job satisfaction, role rejection or withdrawal. Thompson sees BSA affecting the nature of job characteristics for those who engage in this activity by increasing their spheres of action. The following three research hypotheses were generated and tested in this study: (a) BSA will be positively related to role conflict and ambiguity; (b) BSA will be negatively related to job satisfaction; and (c) BSA will be positively related to the job characteristics of variety, autonomy, task identity, feedback, dealings with others, and friendship opportunities.

METHOD

The study was conducted in a large company which manufactures capital equipment via large and small batch technologies. The total sample of 192 (64% response rate) was also broken down by occupational levels as follows: top and middle-level managers (N = 93), engineering personnel (N = 48), and first-level supervisors (N = 33).

Measures

Boundary-Spanning Activity. Prior work [1], [2], [3] had successfully utilized items for the identification of boundary spanners in the technology transfer process in research and development organizations. A four-item BSA scale based on this prior work was developed and validated in an earlier study of boundary-spanning roles in a research and development organization [7]. This BSA scale was used in the present study, with the BSA score being the sum of the four scale items.

To validate the scale for the present sample, management of the manufacturing company was asked to identify some of their employees who management thought were high on BSA. All employees identified by company management as being high on BSA scored in the top third of the BSA scale distribution; moreover, a t-test between the means of the BSA scores of the identified boundary spanners and of the other employees was highly significant ($t = 16.00, p < .001$). The reliability (Coefficient Alpha) of the BSA scale was .81, with an average score of 10.54 and a standard deviation of 3.90.

Role Conflict and Ambiguity. The two scales developed by Rizzo, House, and Lirtzman [12] were used to measure an individual's role perceptions. A factor analysis of the instrument for this sample confirmed the structure of each of the two role variables. In a separate analysis of the data presented in this paper, Szilagyi, Sims, and Keller [15] showed that the role conflict and ambiguity scales had high factor congruencies across multiple samples, as computed by the Wrigley-Neuhaus procedure [5]. The split-half reliabilities (corrected by the Spearman-Brown formula) for this sample were 0.94 for role conflict and .90 for role ambiguity.

Job Satisfaction. The Job Description Index (JDI) was used to measure job satisfaction [14]. The JDI measures satisfaction with five aspects of a job: the work itself, the supervision, the co-workers, the pay, and the opportunities for promotion on the job. The validity and reliability of this instrument have been well established in the literature [14].

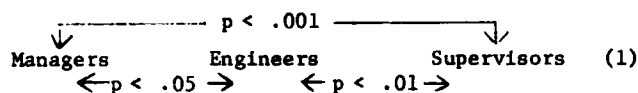
Job Characteristics. Many of the questions in the job characteristics instrument were taken from the Hackman and Lawler [4] research. In order to improve reliability, other questions which appeared to have face validity were added to each scale. A full description of the 35-item instrument, including extensive validity and reliability analyses across multiple samples, can be found in [13]. Overall, the variables exceeded acceptable criteria for validity and reliability. The six job characteristic dimensions measured by the instrument were variety, autonomy, task identity, feedback, dealing with others, and friendship opportunities. Split-half reliabilities (corrected by the Spearman-Brown formula) for the sample in this study were as follows: 0.93 for variety, 0.93 for autonomy, 0.90 for feedback, 0.88 for task identity, 0.89 for dealing, and 0.89 for friendship.

RESULTS

Correlational analyses are reported in Table 1. The analysis for the total sample shows almost no relationships between BSA and role conflict and ambiguity; therefore, the first hypothesis was not supported. BSA was positively related to satisfaction with the work itself and with co-workers. These relationships were opposite to those of hypothesis two; therefore, no support was given for the second hypothesis.

Striking relationships appeared between BSA and the job characteristics. Here, BSA was positively related to variety, autonomy, task identity, friendship opportunities, and dealing with others which were relationships predicted in hypothesis three.

BSA was also analyzed across the three occupational groups by Duncan's multiple range test with Kramer's [8] extension for unequal sample sizes:



It is clear that BSA increases directly with occupational level from supervisors to engineers to managers (Spearman rho = .35, $p < .001$).

Since BSA was significantly different for each occupational level, the data were also analyzed for each level using the three research hypotheses, and are presented in Table 1. For the managerial group, the first two research hypotheses were not supported while the third hypothesis was supported. The same general relationships as those with the managers were found for the engineers with the exception that BSA was negatively related to task identity. The data for the supervisors, however, showed different relationships. Here, BSA was positively related to role conflict, and negatively related to satisfaction with opportunities for promotion. Some small support for the first two hypotheses can therefore be found from the results with the supervisors.

TABLE 1

Zero-Order Correlations Between Boundary-Spanning Activity and Role, Satisfaction, and Job Characteristic Variables (Two-Tailed Tests of Significance)

Variable	Total Sample (N=192)	Managers (N=93)	Engineers (N=48)	Supervisors (N=33)
ROLE VARIABLES				
Role Conflict	.04	.13	-.04	.42*
Role Ambiguity	.08	.04	.18	.06
JOB SATISFACTIONS				
Work	.30***	.24*	.32*	.08
Pay	.05	.11	-.10	-.29
Supervision	.07	.10	.19	-.22
Promotions	.06	.17	.00	-.47**
Co-Workers	.14*	.05	.28*	-.09
JOB CHARACTERISTICS				
Variety	.36***	.45***	.28*	-.25
Autonomy	.23**	.38**	-.08	.09
Task Identity	.14*	.10	-.28*	.32
Feedback	.02	.02	.07	.00
Friendship	.25***	.27**	-.14	.28
Dealing with Others	.14*	.25*	-.17	.22

* $p < .05$
 ** $p < .01$
 *** $p < .001$

DISCUSSION

The results of this study are basically unsupportive of the Kahn et al. [6] theoretical framework and much of the prior research relating to BSA. The total sample results suggest that positions with high levels of BSA are desirable and more satisfying for incumbents.

The data also indicated that BSA is a more important and desirable position component at higher occupational levels. Consistent with Thompson's [16] conceptual framework, BSA increased with an increase in occupational level. At the lower level of supervision, moreover, BSA appears to take on some of the negative relationships hypothesized from the Kahn et al. theory. Here, higher role conflict and lower job satisfaction with opportunities for promotion are related to BSA. One should also note that the number of significant relationships between BSA and the other variables increased with occupation level; hence, BSA appears to be a more important variable for explaining behavioral relationships at higher levels of the organization. This pattern is also consistent with Thompson's framework. The data also suggest, as predicted by Thompson in hypothesis three, that BSA itself may also increase the amount of variety, autonomy, task identity, friendship opportunities, and dealing with others in a job. Much of the prior theory and research relating to BSA has

focused on its negative aspects without fully appreciating the potentially positive aspects that BSA can have for those who engage in this activity.

This study's finding that BSA is positively related to job satisfaction may possibly be explained with the concept of information as a power resource. Pettigrew [11] found that those individuals with high levels of BSA were able to obtain a valuable organizational resource--power, in the form of information. Thompson [16] also suggests that by permitting individuals to exercise discretion, boundary-spanning jobs enable individuals to reduce environmental uncertainties and contingencies, and, to the extent that these aspects are important to the organization, the individual gains power, particularly in a bargaining process. Since those who engage in BSA are able to obtain information from external sources not usually accessible to others, they become more powerful and important to the organization and can use this resource to obtain better and more satisfying jobs.

It may also be that we have been treating BSA as a global variable, and that different types of BSA actually occur in an organization with different consequences. The relative lack of power of lower-level supervisors may increase the conflict and dissatisfaction associated with BSA. Upper-level managers, however, may find that BSA gives them additional power vis-a-vis their peers, and hence it could be a desirable activity.

There is clearly a need for additional research into BSA. Research, perhaps longitudinal, is needed to examine the types of BSA that occur at different levels of an organization. Such research, moreover, should consider the interorganizational context within which the BSA occurs. This perspective is needed to help explain why BSA may have favorable or unfavorable implications for those who engage in the activity.

REFERENCES

- [1] Allen, T. J., and Cohen, S. I. "Information Flow in Research and Development Laboratories," Administrative Science Quarterly, 1969, 14, pp. 12-19.
- [2] Allen, T. J., Piepmeier, J. M., and Cooney, S. "The International Technological Gatekeeper," Technology Review, 1971, p. 73.
- [3] Creighton, J. W., Jolly, J. A., and Denning, S. A. Enhancement of Research and Development Output Utilization Efficiencies: Linker Control Methodology in the Technology Transfer Process. (Monterey, California: Naval Postgraduate School, June 30, 1972).
- [4] Hackman, J. R., and Lawler, E. E. "Employee Reactions to Job Characteristics," Journal of Applied Psychology, 1971, 55, pp. 259-286.
- [5] Harman, H. H. Modern Factor Analysis. (Chicago: University of Chicago Press, 1967).
- [6] Kahn, R. L., Wolfe, D. M., Quinn, R. P., Snoek, J. D., and Rosenthal, R. A. Organizational Stress: Studies in Role Conflict and Ambiguity. (New York: Wiley, 1964).
- [7] Keller, R. T., and Holland, W. E. "Boundary-Spanning Roles in a Research and Development Organization: An Empirical Investigation," Academy of Management Journal, June, 1975.
- [8] Kramer, C. Y. "Extension of Multiple Range Tests to Group Means with Unequal Numbers of Replications," Biometrics, 1956, 12, pp. 307-310.
- [9] Organ, D. W. "Some Variables Affecting Boundary Role Behavior," Sociometry, 1971, 34, pp. 524-537.
- [10] Organ, D. W., and Greene, C. N. "The Boundary Relevance of the Project Manager's Job," R&D Management, 1972, 3, pp. 7-11.
- [11] Pettigrew, A. M. "Information Control as a Power Resource," Sociology, 1972, 6, pp. 187-204.
- [12] Rizzo, J. R., House, R. J., and Lirtzman, S. I. "Role Conflict and Ambiguity in Complex Organizations," Administrative Science Quarterly, 1970, 15, pp. 150-163.
- [13] Sims, H. P., Szilagyi, A. D., and Keller, R. T. "Job Characteristics: Scale Development and Relationships with Satisfaction," Proceedings of the 17th Southwest Academy of Management Annual Meeting, Houston, Texas, March, 1975.
- [14] Smith, P. C., Kendall, L. M., and Hulin, C. L. The Measurement of Satisfaction in Work and Retirement. (Chicago: Rand McNally, 1969).
- [15] Szilagyi, A. D., Sims, H. P., and Keller, R. T. Role Dynamics, Locus of Control, and Employee Attitudes and Behavior. Unpublished manuscript. University of Houston, 1974.
- [16] Thompson, J. D. Organizations in Action. (New York: McGraw-Hill, 1967).
- [17] Wall, J. A., and Adams, J. S. "Some Variables Affecting a Constituent's Evaluations of and Behavior Toward a Boundary Role Occupant," Organizational Behavior and Human Performance, 1974, 11, pp. 390-408.

Copyright of *Academy of Management Proceedings* is the property of *Academy of Management* and its content may not be copied or emailed to multiple sites or posted to a listserv without the copyright holder's express written permission. However, users may print, download, or email articles for individual use.

Copyright of Academy of Management Proceedings (00650668) is the property of Academy of Management and its content may not be copied or emailed to multiple sites or posted to a listserv without the copyright holder's express written permission. However, users may print, download, or email articles for individual use.